Kushal Rana

AI Engineer | Generative AI | Agentic Systems | Backend Architecture

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PROFESSIONAL SUMMARY

AI/ML and backend engineer specializing in **production Generative AI, agentic systems, machine learning, and scalable SaaS platforms**. Experienced in owning systems from architecture and data/model development through production deployment. Architected an AI-powered hospitality Revenue Management System deployed across approximately **70 hotels**, with planned expansion toward **200 properties**, and engineered enterprise RAG and agentic AI infrastructure for complex business workflows.

WORK EXPERIENCE

Aviara Labs **Noida, India**

Software Development Engineer II *Apr 2026 – Present*

AI-Powered Revenue Management System (RMS)

- Led the architecture and development of an end-to-end hospitality Revenue Management System combining **ML-based pricing, hotel demand signals, competitor intelligence, business rules, and automated revenue workflows**.
- Developed end-to-end ML pipelines using approximately **18 months of historical hotel data**, covering data ingestion, preprocessing, feature engineering, model training, evaluation, prediction pipelines, and continuous model improvement.
- Productionized ML pricing intelligence through scalable backend services and REST APIs, integrating model outputs directly into operational hotel revenue-management workflows.
- Designed the platform as a **multi-tenant, multi-property SaaS system** supporting property-level pricing and revenue intelligence; deployed across approximately **70 hotels in Malaysia**, with planned expansion toward **200 properties**.

COVA – Enterprise Agentic AI Infrastructure

- Architected the Generative AI and agentic intelligence layer powering Aviara Connect across contracts, purchase orders, invoices, obligations, renewals, opportunities, partners, and document workflows.
- Built production **RAG and agentic AI workflows** combining semantic and vector retrieval, structured database retrieval, hybrid search, tool calling, specialized agents, conversation context, enterprise memory, and permission-aware data access.
- Designed retrieval and reasoning workflows enabling AI agents to operate across structured enterprise records and unstructured documents while maintaining **tenant isolation and access-control boundaries**.
- Engineered production AI orchestration and backend components using Python, FastAPI, Django, PostgreSQL, MongoDB, Redis, LangChain, LlamaIndex, Docker, Azure, and REST APIs, with emphasis on **scalable data access, reliability, LLM orchestration, and maintainable enterprise architecture**.

Aviara Eye – LLMOps & AI Observability

- Established centralized **LLM observability and prompt-management infrastructure** using a self-hosted Langfuse deployment across production Generative AI applications.
- Instrumented **LLM calls, prompts, tool executions, agent workflows, model interactions, and execution traces**, enabling engineering teams to debug failures, inspect agent behavior, manage prompts centrally, and monitor production AI systems.

Software Development Engineer I *Aug 2024 – Apr 2026*

- Developed and productionized **machine-learning models, backend services, data pipelines, database schemas, APIs, and AI components** used across Aviara Labs' production platforms.
- Progressed from feature-level implementation to ownership of **AI components, backend systems, system design, and production architecture**.
- Built machine-learning, deep-learning, document-processing, retrieval, and Generative AI solutions while implementing reproducible development and deployment workflows using **Docker, Git, Azure, MLflow, DVC, and CI/CD**.

Rapid Autobots **Kerala, India**

Associate AI Developer *Jan 2024 – Jul 2024*

- Developed an AI-powered conversational platform for a drop-shipping e-commerce system using **OpenAI APIs, Python, FastAPI, LangChain, SQL, and MongoDB**.
- Built contextual AI workflows enabling access to **product, order, store, and customer information** from client databases and backend systems.
- Designed backend integrations, automated reporting and Excel export workflows, and containerized services using **Docker and cloud infrastructure**.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++
Generative AI: LLMs, RAG, Agentic AI, Multi-Agent Systems, Hybrid Search, Semantic Search, Tool Calling, Prompt Engineering, LangChain, LlamaIndex, Vector Databases
Machine Learning: Machine Learning, Deep Learning, PyTorch, TensorFlow, Feature Engineering, Model Training, Model Evaluation, ML Pipelines, MLflow, DVC
Backend Engineering: FastAPI, Django, REST APIs, Microservices, API Design
Databases & Data Systems: PostgreSQL, MongoDB, Redis, Vector Databases
LLMOps & AI Observability: Langfuse/LangSmith, LLM Tracing, Prompt Management, Agent Observability, LLM evaluation, AI Application Monitoring
Cloud / DevOps: Microsoft Azure, Docker, Kubernetes, GitHub Actions, CI/CD, Airflow, Linux, Git
System Design: Multi-Tenant SaaS, AI System Architecture, LLM Orchestration, Retrieval Architecture, Scalable Backend Systems

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate — View Credential
- Machine Learning Specialization – DeepLearning.AI — View Credential
- System Design Masterclass (2025) — View Credential
- Docker Mastery: Kubernetes + Docker Swarm — View Credential

EDUCATION

I.K. Gujral Punjab Technical University — **Punjab, India**
Bachelor of Technology in Computer Science and Engineering — *Sep 2020 – Jul 2024*

– **CGPA: 8.0/10**

ACHIEVEMENTS

- Secured **3rd place among 60+ teams** at the Bharat Tech Experience Hackathon in 2024.